

Week 5 - Multi-Model CPU Inference Sweep: Llama-3.2-1B vs. Llama-3.2-3B

Lenovo IdeaPad Flex 5 (Intel i7, no GPU), llama.cpp, RAPL package-0, governor = performance. Both models Q4_K_M. Same bubble-sort task and padding as week 3. Colour key used throughout: blue = Llama-3.2-1B, orange = Llama-3.2-3B.

1. Experiment 1 - Prefill/Decode Scaling

Input sweep: 8 input settings (nominal 128-2048), output fixed at 64. Output sweep: 8 output settings (nominal 8-512), input fixed at 512 nominal (422 actual). 5 repeats each, 80 runs per model. Token counts are actual counts from llama-tokenize.

1.1 Raw Data - Llama-3.2-1B (80 runs; 8 input settings x 5 repeats, 8 output settings x 5 repeats)

sweep	input_req	output_req	run	input_act	output_act	energy (J)	wall time (s)
input	128	64	1	104	64	108.0142	5.0362
input	128	64	2	104	64	109.3231	5.0373
input	128	64	3	104	64	108.6582	5.0364
input	128	64	4	104	64	109.6378	5.0370
input	128	64	5	104	64	107.9470	5.0370
input	256	64	1	211	64	133.4521	6.0436
input	256	64	2	211	64	133.4425	6.0429
input	256	64	3	211	64	134.4983	6.0442
input	256	64	4	211	64	138.4836	6.0434
input	256	64	5	211	64	134.9169	6.0432
input	384	64	1	316	64	159.5390	7.0511
input	384	64	2	316	64	158.3169	7.0517
input	384	64	3	316	64	227.1931	8.0583
input	384	64	4	316	64	159.2254	7.0521
input	384	64	5	316	64	182.3507	7.0493
input	512	64	1	422	64	189.5060	8.0580
input	512	64	2	422	64	189.1917	8.0586
input	512	64	3	422	64	187.4116	8.0587
input	512	64	4	422	64	189.4417	8.0587
input	512	64	5	422	64	186.4796	8.0591
input	768	64	1	635	64	242.0843	10.0730
input	768	64	2	635	64	240.5207	10.0730
input	768	64	3	635	64	243.3050	10.0726
input	768	64	4	635	64	241.4410	10.0727
input	768	64	5	635	64	240.8602	10.0717
input	1024	64	1	846	64	296.4441	12.0867
input	1024	64	2	846	64	295.3553	12.0877
input	1024	64	3	846	64	299.9649	12.0867
input	1024	64	4	846	64	300.0059	12.0882
input	1024	64	5	846	64	297.0829	12.0862
input	1536	64	1	1270	64	413.0787	16.1130
input	1536	64	2	1270	64	414.6260	16.1144
input	1536	64	3	1270	64	418.9730	17.1214
input	1536	64	4	1270	64	420.6832	17.1239
input	1536	64	5	1270	64	416.3194	16.1159
input	2048	64	1	1693	64	444.9435	24.2146
input	2048	64	2	1693	64	533.7962	21.1525
input	2048	64	3	1693	64	437.8675	24.2162
input	2048	64	4	1693	64	531.7066	21.1562
input	2048	64	5	1693	64	427.7830	24.2215
output	512	8	1	422	8	135.4383	6.0450
output	512	8	2	422	8	134.0891	6.0447
output	512	8	3	422	8	134.6323	6.0444

sweep	input_req	output_req	run	input_act	output_act	energy (J)	wall time (s)
output	512	8	4	422	8	135.0559	6.0443
output	512	8	5	422	8	134.6977	6.0447
output	512	16	1	422	16	141.9386	6.0433
output	512	16	2	422	16	148.7177	6.0435
output	512	16	3	422	16	142.4727	6.0449
output	512	16	4	422	16	140.4783	6.0444
output	512	16	5	422	16	140.3987	6.0441
output	512	32	1	422	32	157.3117	7.0525
output	512	32	2	422	32	157.1257	7.0517
output	512	32	3	422	32	157.3858	7.0518
output	512	32	4	422	32	158.3855	7.0515
output	512	32	5	422	32	157.1618	7.0525
output	512	64	1	422	64	185.2362	8.0591
output	512	64	2	422	64	187.9490	8.0606
output	512	64	3	422	64	186.6243	8.0594
output	512	64	4	422	64	189.2774	8.0587
output	512	64	5	422	64	191.7706	8.0590
output	512	128	1	422	128	246.8380	11.0811
output	512	128	2	422	128	244.4306	10.0744
output	512	128	3	422	128	251.8710	11.0835
output	512	128	4	422	128	248.1654	11.0807
output	512	128	5	422	128	244.9533	10.0744
output	512	256	1	422	255	370.0512	16.1179
output	512	256	2	422	256	368.9563	16.1171
output	512	256	3	422	256	362.5388	15.1121
output	512	256	4	422	255	359.8073	15.1131
output	512	256	5	422	256	360.8550	15.1128
output	512	384	1	422	374	489.6272	21.1548
output	512	384	2	422	384	521.4772	21.1529
output	512	384	3	422	372	440.1889	22.1838
output	512	384	4	422	332	440.6789	19.1393
output	512	384	5	422	384	470.1552	21.1646
output	512	512	1	422	377	408.7015	22.1933
output	512	512	2	422	466	552.7350	25.1953
output	512	512	3	422	418	449.7062	25.2199
output	512	512	4	422	342	445.3741	19.1396
output	512	512	5	422	404	436.4271	23.2029

1.2 Raw Data - Llama-3.2-3B (80 runs; 8 input settings x 5 repeats, 8 output settings x 5 repeats)

sweep	input_req	output_req	run	input_act	output_act	energy (J)	wall time (s)
input	128	64	1	104	64	257.6380	12.0890
input	128	64	2	104	64	271.2884	12.0848
input	128	64	3	104	64	268.8840	12.0861
input	128	64	4	104	64	274.8280	12.0859
input	128	64	5	104	64	269.1631	12.0855
input	256	64	1	211	64	344.7943	14.0992
input	256	64	2	211	64	342.5864	14.1000
input	256	64	3	211	64	325.1277	15.1177
input	256	64	4	211	64	306.7349	15.1244
input	256	64	5	211	64	346.7899	14.1011
input	384	64	1	316	64	369.0186	19.1596
input	384	64	2	316	64	421.2633	17.1216
input	384	64	3	316	64	367.4803	19.1629
input	384	64	4	316	64	416.7025	18.1324
input	384	64	5	316	64	361.7618	19.1665

sweep	input_req	output_req	run	input_act	output_act	energy (J)	wall time (s)
input	512	64	1	422	64	477.4546	20.1534
input	512	64	2	422	64	412.1954	23.2047
input	512	64	3	422	64	477.7523	21.1636
input	512	64	4	422	64	410.2151	22.1947
input	512	64	5	422	64	452.2038	21.1711
input	768	64	1	635	64	527.7439	30.2790
input	768	64	2	635	64	585.3596	28.2367
input	768	64	3	635	64	539.6098	31.2824
input	768	64	4	635	64	590.2274	28.2347
input	768	64	5	635	64	529.8584	30.2763
input	1024	64	1	846	64	698.4790	36.3130
input	1024	64	2	846	64	647.0879	38.3515
input	1024	64	3	846	64	712.2832	35.3030
input	1024	64	4	846	64	644.6312	38.3552
input	1024	64	5	846	64	696.4837	36.3185
input	1536	64	1	1270	64	878.7827	53.5036
input	1536	64	2	1270	64	941.1180	51.4644
input	1536	64	3	1270	64	885.2123	54.5117
input	1536	64	4	1270	64	949.0722	51.4620
input	1536	64	5	1270	64	895.0362	54.5228
input	2048	64	1	1693	64	1114.8925	69.6579
input	2048	64	2	1693	64	1115.2926	69.6700
input	2048	64	3	1693	64	1110.0405	68.6567
input	2048	64	4	1693	64	1136.8086	70.6787
input	2048	64	5	1693	64	1106.1424	68.6595
output	512	8	1	422	8	319.5044	16.1368
output	512	8	2	422	8	317.6353	16.1400
output	512	8	3	422	8	317.6971	16.1398
output	512	8	4	422	7	319.6239	16.1412
output	512	8	5	422	7	361.2643	14.1011
output	512	16	1	422	16	330.0079	17.1497
output	512	16	2	422	16	332.4723	17.1519
output	512	16	3	422	16	332.0801	17.1471
output	512	16	4	422	16	377.4067	16.1192
output	512	16	5	422	16	333.7820	17.1502
output	512	32	1	422	32	357.7424	19.1675
output	512	32	2	422	32	396.2902	17.1353
output	512	32	3	422	32	357.6894	19.1702
output	512	32	4	422	32	359.3537	19.1687
output	512	32	5	422	32	409.3707	17.1290
output	512	64	1	422	64	406.6967	22.1993
output	512	64	2	422	64	411.8858	22.2040
output	512	64	3	422	64	410.6463	22.1995
output	512	64	4	422	64	409.1680	22.2010
output	512	64	5	422	64	410.9397	22.2007
output	512	128	1	422	128	579.6755	29.2449
output	512	128	2	422	128	535.5847	30.2749
output	512	128	3	422	128	530.2861	30.2765
output	512	128	4	422	128	531.9500	30.2765
output	512	128	5	422	128	522.5658	31.2844
output	512	256	1	422	255	772.5694	46.4186
output	512	256	2	422	256	769.5917	46.4235
output	512	256	3	422	256	774.4793	46.4226
output	512	256	4	422	255	770.9011	46.4244
output	512	256	5	422	256	787.8601	47.4430

sweep	input_req	output_req	run	input_act	output_act	energy (J)	wall time (s)
output	512	384	1	422	384	1021.2249	63.5849
output	512	384	2	422	384	1017.2600	62.5772
output	512	384	3	422	384	1038.6481	64.6093
output	512	384	4	422	369	1000.0764	61.5685
output	512	384	5	422	384	1025.7042	63.5862
output	512	512	1	422	454	1167.4752	72.6729
output	512	512	2	422	510	1292.6368	81.7566
output	512	512	3	422	461	1179.4868	73.6800
output	512	512	4	422	448	1139.2692	70.6510
output	512	512	5	422	453	1164.1813	72.6714

1.3 Regression Statistics and Constant Overhead

	slope (J/token)	intercept - constant overhead (J)	R-squared
Llama-3.2-1B, output sweep (decode)	0.8747	131.42	0.9817
Llama-3.2-1B, input sweep (prefill)	0.2378	91.72	0.9674
Llama-3.2-3B, output sweep (decode)	1.8801	305.20	0.9969
Llama-3.2-3B, input sweep (prefill)	0.5364	218.38	0.9925
Week 3 reference: Llama-3.2-1B, Run B decode	0.7591	167.74	0.9978
Week 3 reference: Llama-3.2-1B, Run B prefill	0.2000	144.39	0.9850

	Llama-3.2-1B	Llama-3.2-3B	3B / 1B
Decode marginal cost	0.8747 J/token	1.8801 J/token	2.15x
Prefill marginal cost	0.2378 J/token	0.5364 J/token	2.26x
Decode / Prefill ratio	3.68	3.50	-
Constant overhead (decode-fit intercept)	131.4 J	305.2 J	2.32x
Model file size (Q4_K_M)	771 MB	1.9 GB	2.5x

Slopes are fitted on actual token counts across all 40 runs of each sweep. Week 3 rows are from fyp_print_data (5 input / 6 output settings, 3 repeats) and are shown for continuity only.

1.4 Decode Share of Total Energy vs. Output Length

output_req	1B actual output	1B measured avg (J)	1B predicted decode (J)	1B decode %	3B actual output	3B measured avg (J)	3B predicted decode (J)	3B decode %
8	8.0	134.8	7.0	5.2%	7.6	327.1	14.3	4.4%
16	16.0	142.8	14.0	9.8%	16.0	341.1	30.1	8.8%
32	32.0	157.5	28.0	17.8%	32.0	376.1	60.2	16.0%
64	64.0	188.2	56.0	29.8%	64.0	409.9	120.3	29.4%
128	128.0	247.3	112.0	45.3%	128.0	540.0	240.6	44.6%
256	255.6	364.4	223.6	61.3%	255.6	775.1	480.5	62.0%
384	369.2	472.4	323.0	68.4%	381.0	1020.6	716.3	70.2%
512	401.4	458.6	351.1	76.6%	465.2	1188.6	874.6	73.6%

Predicted decode energy = decode slope x actual output tokens (mean of 5 runs). At the 384 and 512 budgets the models stop early, so actual output is below the request.

1.5 Inter-Model Comparison per Setting (means of 5 runs)

sweep	setting	1B energy (J)	3B energy (J)	3B / 1B energy	1B avg power (W)	3B avg power (W)	1B decode (tok/s)	3B decode (tok/s)	1B prefill (tok/s)	3B prefill (tok/s)
input	in 128	108.7	268.4	2.47x	21.6	22.2	27.0	10.6	117.8	40.5
input	in 256	135.0	333.2	2.47x	22.3	23.0	25.6	9.9	118.7	41.1
input	in 384	177.3	387.2	2.18x	24.3	21.0	25.6	9.4	114.0	38.5
input	in 512	188.4	446.0	2.37x	23.4	20.8	25.6	8.9	116.2	38.2
input	in 768	241.6	554.6	2.29x	24.0	18.8	24.6	7.9	116.1	35.0
input	in 1024	297.8	679.8	2.28x	24.6	18.5	23.3	7.4	113.3	34.3
input	in 1536	416.7	909.8	2.18x	25.2	17.2	19.7	6.6	110.8	31.8
input	in 2048	475.2	1116.6	2.35x	20.9	16.1	16.3	6.1	98.5	30.4

sweep	setting	1B energy (J)	3B energy (J)	3B / 1B energy	1B avg power (W)	3B avg power (W)	1B decode (tok/s)	3B decode (tok/s)	1B prefill (tok/s)	3B prefill (tok/s)
output	out 8	134.8	327.1	2.43x	22.3	20.9	25.7	8.8	117.0	36.2
output	out 16	142.8	341.1	2.39x	23.6	20.2	25.7	8.9	115.8	36.0
output	out 32	157.5	376.1	2.39x	22.3	20.6	25.8	8.9	115.8	37.1
output	out 64	188.2	409.9	2.18x	23.3	18.5	25.3	8.8	116.1	35.1
output	out 128	247.3	540.0	2.18x	23.2	17.9	24.9	8.3	116.9	35.8
output	out 256	364.4	775.1	2.13x	23.5	16.6	25.6	8.1	116.6	35.1
output	out 384	472.4	1020.6	2.16x	22.6	16.2	24.1	8.0	117.5	34.6
output	out 512	458.6	1188.6	2.59x	20.1	16.0	23.1	7.9	117.4	35.0

Average power = energy / wall time per run. Decode and prefill speeds are llama-cli's own reported throughput.

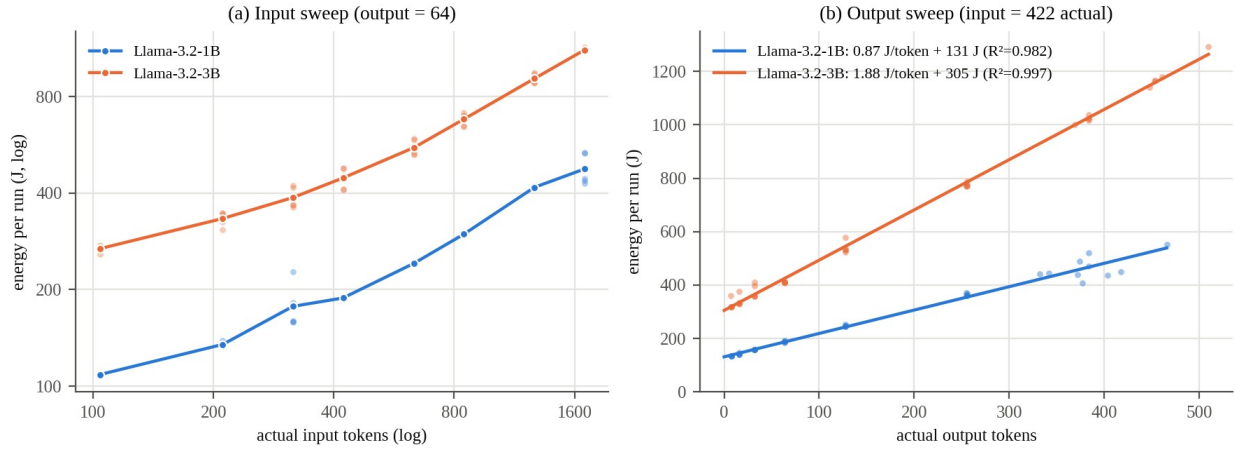


Figure B1. Energy vs. input length (a, log-log) and output length with linear fit (b), both models (80 runs each). Faint dots are individual runs, solid markers are means.

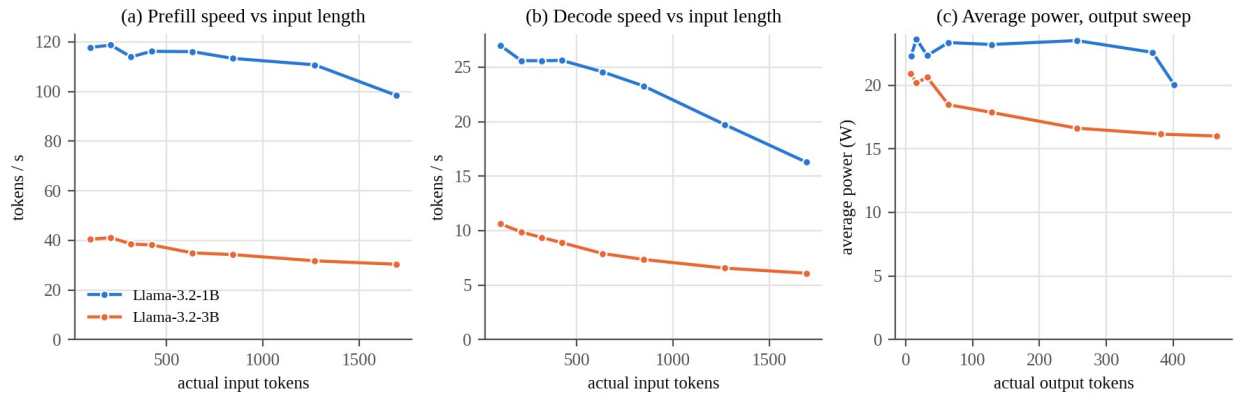


Figure B2. Prefill speed (a) and decode speed (b) vs. input length, and average power draw across the output sweep (c), both models.

CPU (core 0) frequency vs. elapsed time, all repeats, input sweep, Llama-3.2-1B

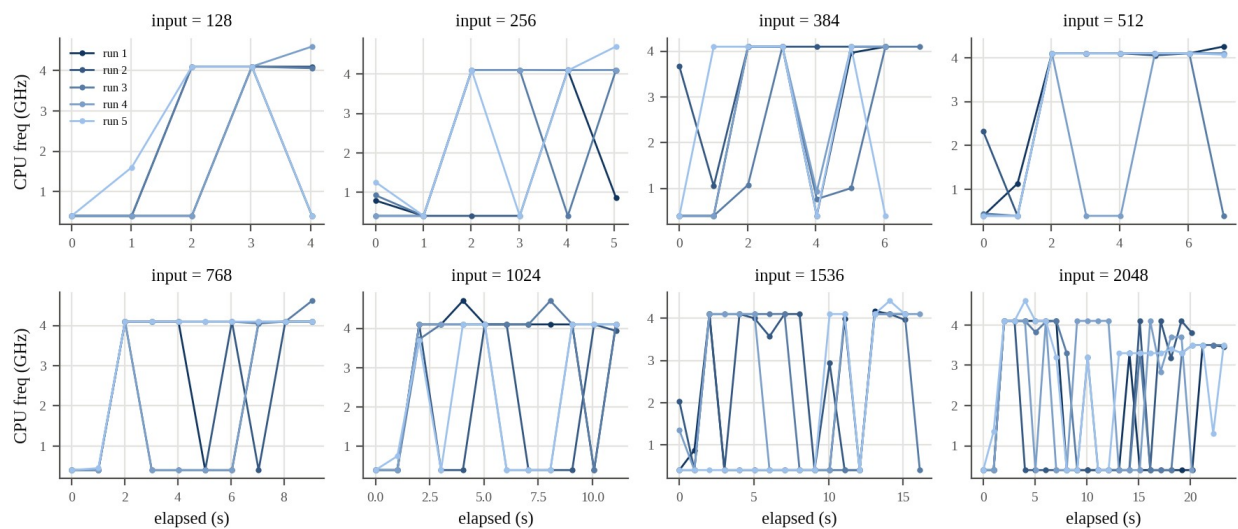


Figure A1. CPU (core 0) frequency vs. elapsed time, all repeats, input sweep, Llama-3.2-1B.

CPU (core 0) frequency vs. elapsed time, all repeats, output sweep, Llama-3.2-1B

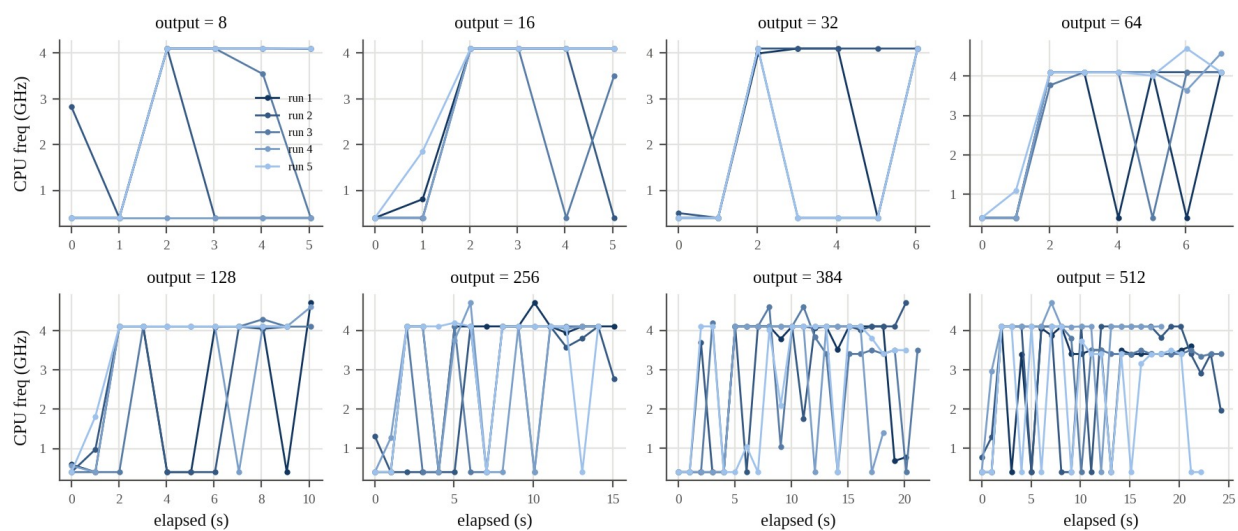


Figure A2. CPU (core 0) frequency vs. elapsed time, all repeats, output sweep, Llama-3.2-1B.

CPU (core 0) frequency vs. elapsed time, all repeats, input sweep, Llama-3.2-3B

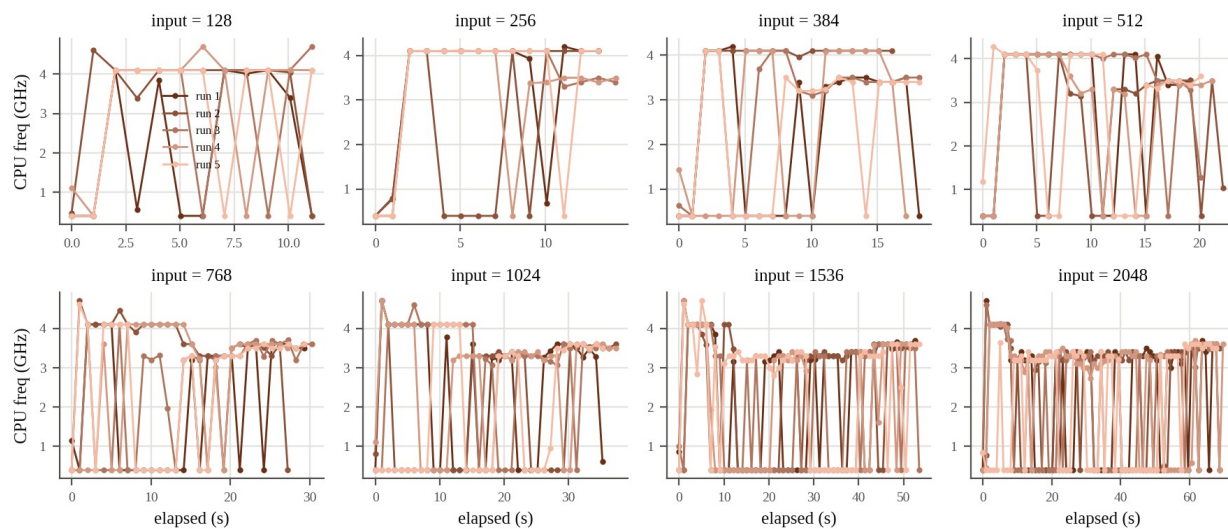


Figure A3. CPU (core 0) frequency vs. elapsed time, all repeats, input sweep, Llama-3.2-3B.

CPU (core 0) frequency vs. elapsed time, all repeats, output sweep, Llama-3.2-3B

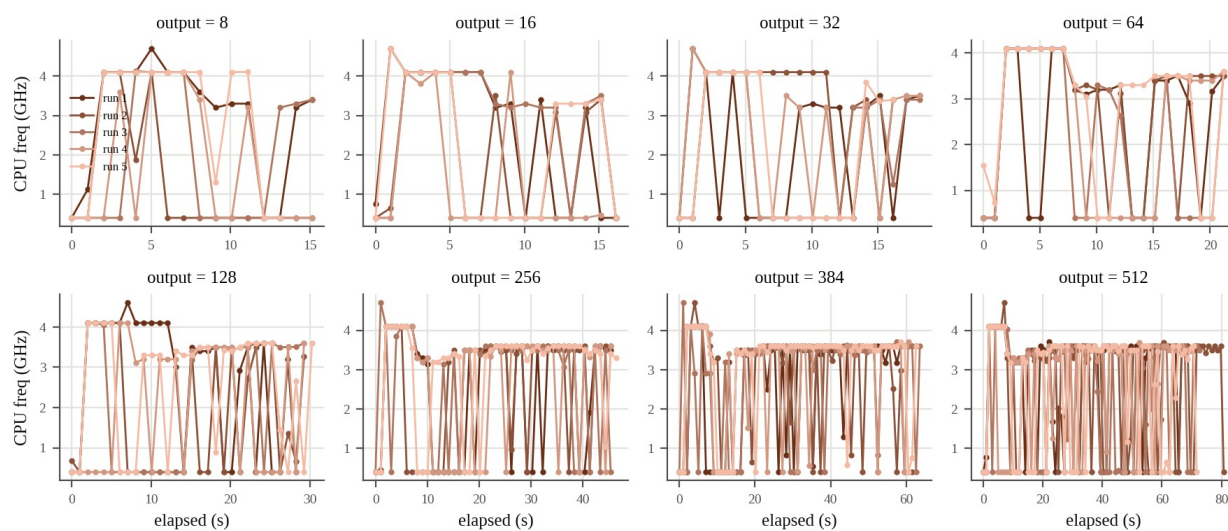


Figure A4. CPU (core 0) frequency vs. elapsed time, all repeats, output sweep, Llama-3.2-3B.

Cumulative energy vs. elapsed time, all repeats, input sweep, Llama-3.2-1B

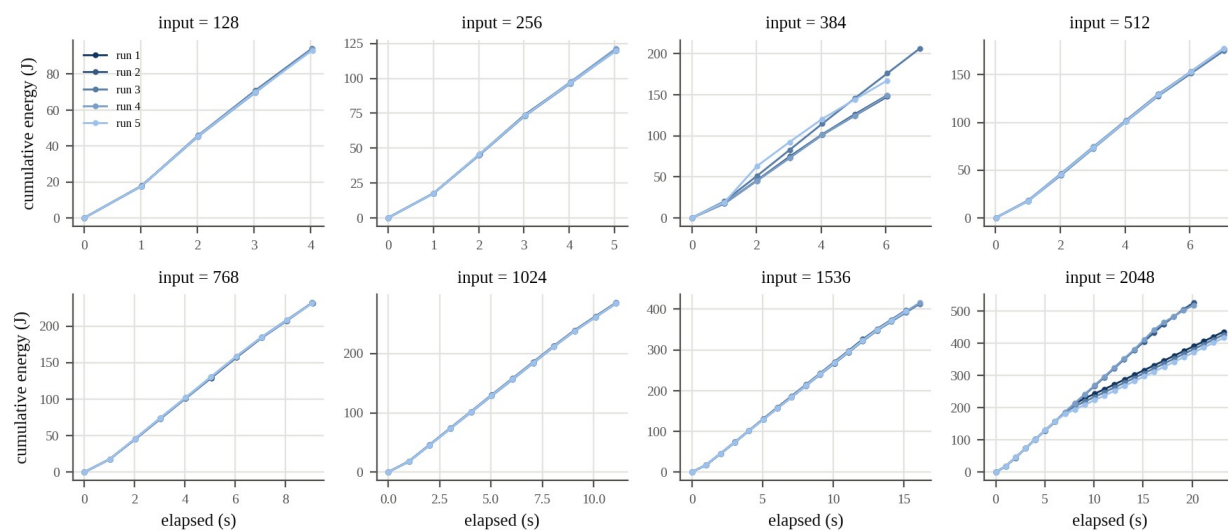


Figure A5. Cumulative energy vs. elapsed time, all repeats, input sweep, Llama-3.2-1B.

Cumulative energy vs. elapsed time, all repeats, output sweep, Llama-3.2-1B

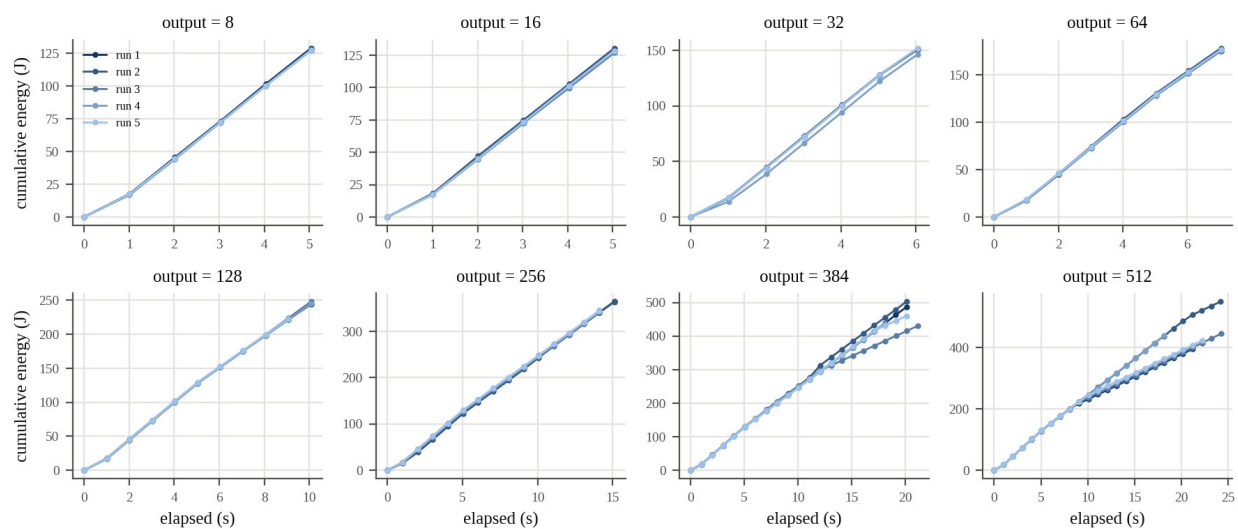


Figure A6. Cumulative energy vs. elapsed time, all repeats, output sweep, Llama-3.2-1B.

Cumulative energy vs. elapsed time, all repeats, input sweep, Llama-3.2-3B

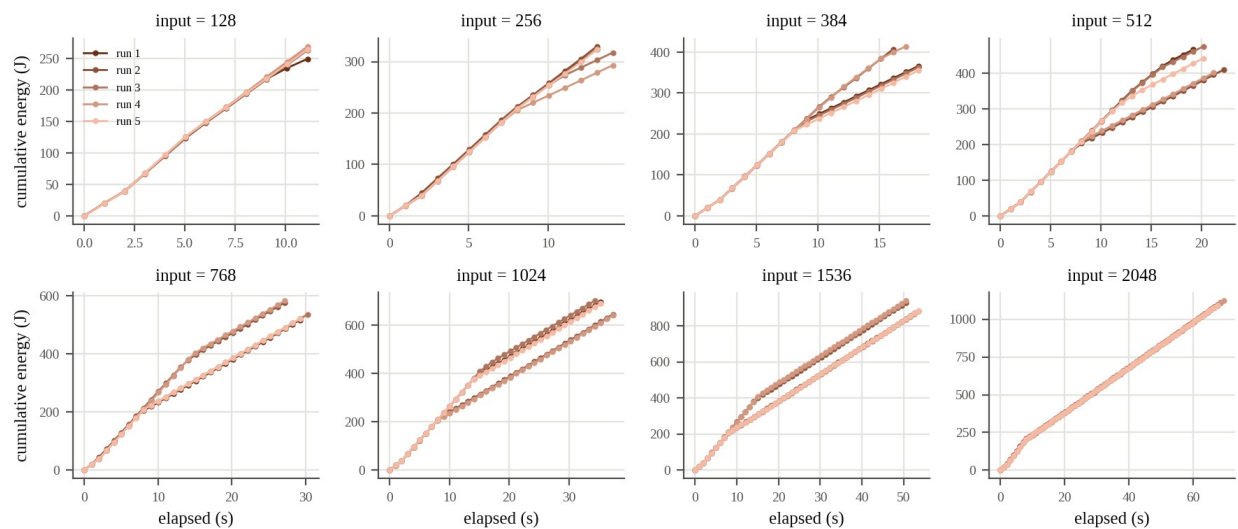


Figure A7. Cumulative energy vs. elapsed time, all repeats, input sweep, Llama-3.2-3B.

Cumulative energy vs. elapsed time, all repeats, output sweep, Llama-3.2-3B

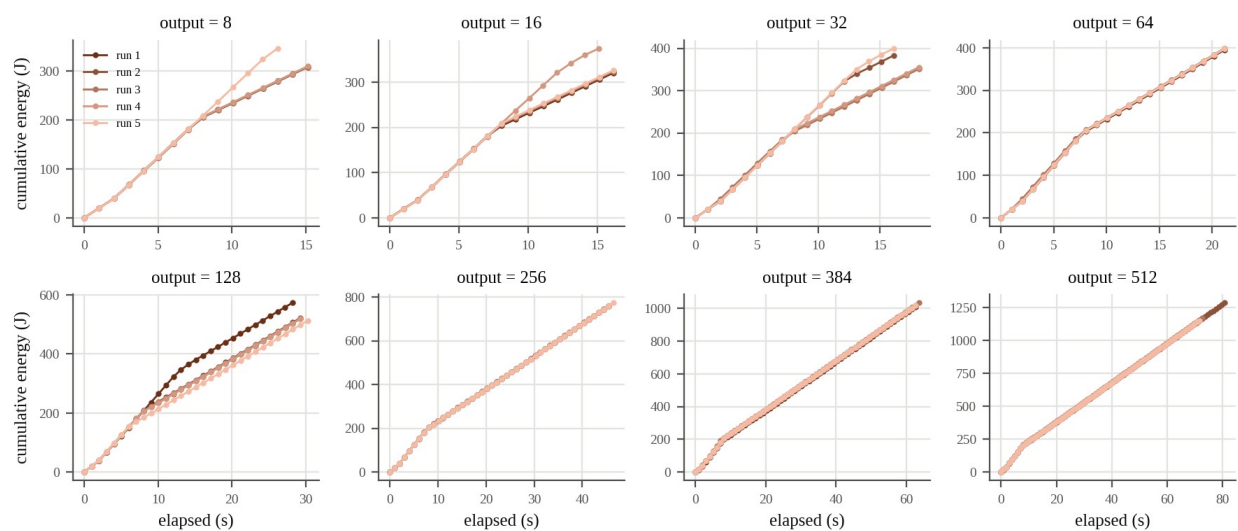


Figure A8. Cumulative energy vs. elapsed time, all repeats, output sweep, Llama-3.2-3B.

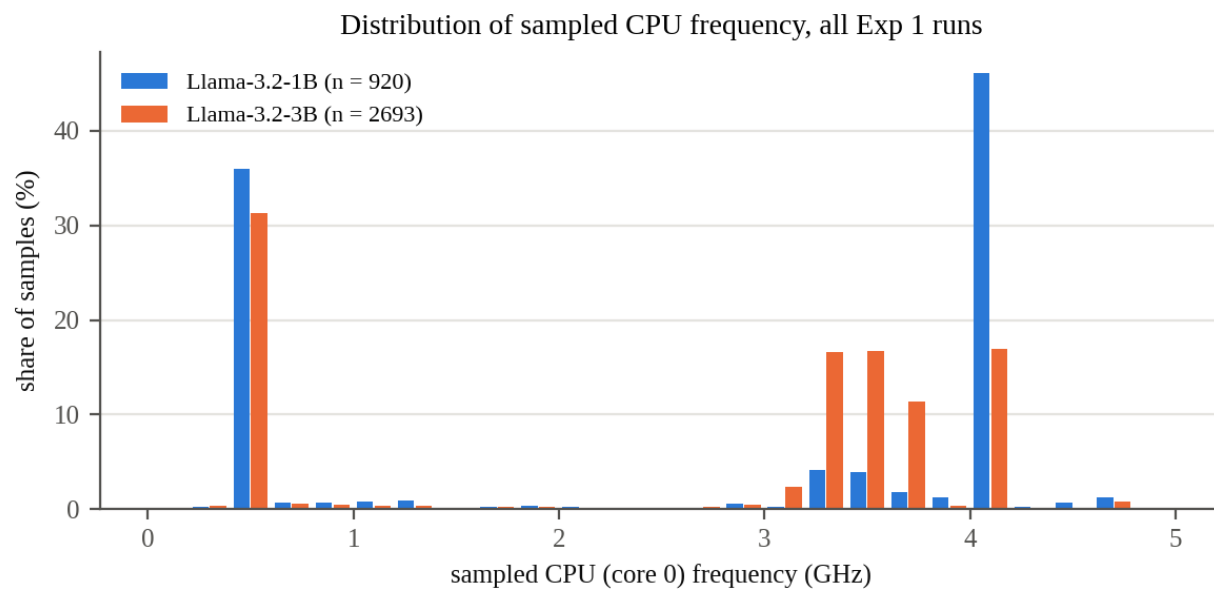


Figure A9. Distribution of sampled CPU (core 0) frequency as a share of samples, all Exp 1 runs and settings (n = 920 samples for 1B, 2693 for 3B).

2. Experiment 2 - Decoding Strategy Comparison

Fixed input (422 actual tokens), 256-token output budget. Strategies: greedy (temp=0), temperature (temp=0.8), top-p / nucleus (temp=0.8, top-p=0.9), top-k (temp=0.8, top-k=40). Sampler defaults explicitly disabled in conditions not testing them. 5 repeats per strategy per model.

2.1 Raw Data (40 runs)

model	strategy	run	output tokens	energy (J)	wall time (s)	avg power (W)	working fn (harness)	working fn (re-scored)
Llama-3.2-1B	greedy	1	256	364.2850	15.1039	24.12	No	Yes
Llama-3.2-1B	greedy	2	256	368.0686	16.1137	22.84	No	Yes
Llama-3.2-1B	greedy	3	256	369.4711	16.1152	22.93	No	Yes
Llama-3.2-1B	greedy	4	256	383.6973	16.1136	23.81	No	Yes
Llama-3.2-1B	greedy	5	256	373.6014	16.1142	23.18	No	Yes
Llama-3.2-1B	temperature	1	256	373.1240	16.1142	23.15	No	Yes
Llama-3.2-1B	temperature	2	256	422.0635	16.1118	26.20	Yes	Yes
Llama-3.2-1B	temperature	3	256	367.4245	16.1157	22.80	Yes	Yes
Llama-3.2-1B	temperature	4	256	369.1231	16.1162	22.90	No	Yes
Llama-3.2-1B	temperature	5	256	379.1009	16.1154	23.52	No	Yes
Llama-3.2-1B	top_p	1	255	371.3917	16.1179	23.04	Yes	Yes
Llama-3.2-1B	top_p	2	256	369.9975	16.1168	22.96	No	Yes
Llama-3.2-1B	top_p	3	256	368.1913	16.1148	22.85	Yes	Yes
Llama-3.2-1B	top_p	4	256	364.6174	15.1099	24.13	Yes	Yes
Llama-3.2-1B	top_p	5	256	377.2487	16.1160	23.41	No	Yes
Llama-3.2-1B	top_k	1	256	368.5588	16.1158	22.87	Yes	Yes
Llama-3.2-1B	top_k	2	256	371.6624	16.1153	23.06	Yes	Yes
Llama-3.2-1B	top_k	3	255	364.0262	15.1104	24.09	Yes	Yes
Llama-3.2-1B	top_k	4	256	369.4202	16.1162	22.92	No	No
Llama-3.2-1B	top_k	5	256	364.4463	15.1079	24.12	No	Yes
Llama-3.2-3B	greedy	1	256	864.0828	43.3355	19.94	No	Yes
Llama-3.2-3B	greedy	2	256	838.2815	43.3549	19.34	No	Yes
Llama-3.2-3B	greedy	3	256	841.4908	44.3665	18.97	No	Yes
Llama-3.2-3B	greedy	4	256	845.4413	44.3654	19.06	No	Yes
Llama-3.2-3B	greedy	5	256	836.8921	44.3704	18.86	No	Yes
Llama-3.2-3B	temperature	1	256	833.1965	44.3691	18.78	No	Yes
Llama-3.2-3B	temperature	2	256	819.8143	45.3866	18.06	Yes	Yes
Llama-3.2-3B	temperature	3	256	845.5659	45.3809	18.63	Yes	Yes
Llama-3.2-3B	temperature	4	256	832.1597	44.3749	18.75	Yes	Yes
Llama-3.2-3B	temperature	5	255	831.5987	44.3717	18.74	No	No
Llama-3.2-3B	top_p	1	256	773.8200	46.4188	16.67	No	No
Llama-3.2-3B	top_p	2	256	853.9860	44.3690	19.25	No	Yes
Llama-3.2-3B	top_p	3	256	818.6498	45.3898	18.04	No	No
Llama-3.2-3B	top_p	4	256	828.8241	44.3689	18.68	Yes	Yes
Llama-3.2-3B	top_p	5	256	811.3112	45.3911	17.87	No	Yes
Llama-3.2-3B	top_k	1	256	811.3129	44.3842	18.28	Yes	Yes
Llama-3.2-3B	top_k	2	256	769.1924	46.4121	16.57	Yes	Yes
Llama-3.2-3B	top_k	3	256	841.3791	44.3689	18.96	Yes	Yes
Llama-3.2-3B	top_k	4	256	769.7257	46.4148	16.58	No	No
Llama-3.2-3B	top_k	5	256	842.1445	44.3689	18.98	Yes	Yes

"Working function" replaces week 3's "reached return?": the generated code is extracted and the function must correctly sort 7 test lists. "Harness" is the check recorded during the run (scripts from commit 7d8b567, unchanged through eed365e), which fails any output cut off mid example-usage line even when the function above it is complete. "Re-scored" is scripts/rescore_code.py on the saved outputs (results/*all_models_rescored.csv, column passes_tests_v2): it drops the unparseable trailing lines first, so a function that is itself cut off still fails. Summaries and Figure C1 use the re-scored column.

2.2 Summary Statistics by Strategy and Model

strategy	1B avg energy (J)	1B avg power (W)	1B working (harness / re-scored)	3B avg energy (J)	3B avg power (W)	3B working (harness / re-scored)	3B / 1B energy
greedy	371.8 $\pm$ 7.4	23.38	0 / 5 of 5	845.2 $\pm$ 11.0	19.23	0 / 5 of 5	2.27x
temperature	382.2 $\pm$ 22.8	23.72	2 / 5 of 5	832.5 $\pm$ 9.1	18.59	3 / 4 of 5	2.18x
top_p	370.3 $\pm$ 4.6	23.28	3 / 5 of 5	817.3 $\pm$ 29.2	18.10	1 / 3 of 5	2.21x
top_k	367.6 $\pm$ 3.3	23.41	3 / 4 of 5	806.8 $\pm$ 36.2	17.88	4 / 4 of 5	2.19x

$\pm$ is the standard deviation across the 5 runs. Every run of both models used the full 256-token budget.

Decoding strategy comparison, 256-token output budget, input 422 tokens

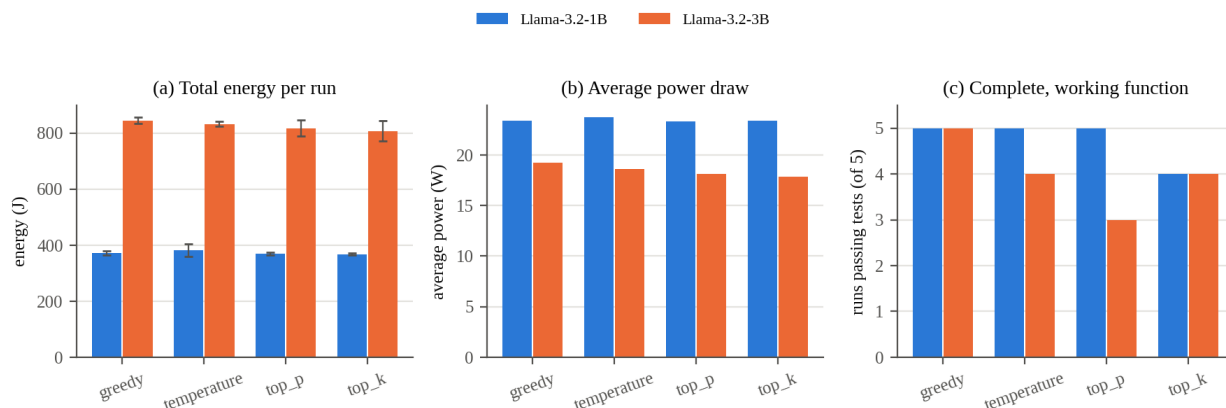


Figure C1. Total energy with ± 1 SD (a), average power draw (b), and runs producing a complete, working function (c) by decoding strategy, both models.

3. Experiment 3 - Energy per Completed Task (new)

Bare task prompt ("Write a function to sort a list of numbers using bubble sort.", 13 tokens, no padding). Output cap 1024 tokens; the model stops when it decides it is done. Same sampling for both models (temp=0.6, top-p=0.95). 5 runs per model.

3.1 Raw Data (10 runs)

model	run	output tokens	energy (J)	wall time (s)	avg power (W)	decode (tok/s)	working fn (harness)	working fn (re-scored)
Llama-3.2-1B	1	587	585.1587	25.1719	23.25	25.1	Yes	Yes
Llama-3.2-1B	2	371	358.6411	16.1121	22.26	27.2	Yes	Yes
Llama-3.2-1B	3	521	511.6529	22.1575	23.09	25.5	Yes	Yes
Llama-3.2-1B	4	450	430.3362	19.1379	22.49	27.1	Yes	Yes
Llama-3.2-1B	5	488	481.3652	21.1549	22.75	26.0	Yes	Yes
Llama-3.2-3B	1	557	1330.3389	58.4003	22.78	10.2	Yes	Yes
Llama-3.2-3B	2	503	1131.3655	51.3890	22.02	10.4	Yes	Yes
Llama-3.2-3B	3	574	1193.1955	62.5074	19.09	9.7	Yes	Yes
Llama-3.2-3B	4	447	974.5343	48.3793	20.14	9.9	Yes	Yes
Llama-3.2-3B	5	514	1068.9811	59.4914	17.97	9.2	Yes	Yes

3.2 Summary Statistics

	Llama-3.2-1B	Llama-3.2-3B	3B / 1B
Energy per task (J), mean $\pm$ SD	473.4 $\pm$ 85.2	1139.7 $\pm$ 133.7	2.41x
Output tokens, mean $\pm$ SD	483.4 $\pm$ 80.5	519.0 $\pm$ 49.8	1.07x
Energy per output token (J)	0.978	2.195	2.25x
Wall time per task (s)	20.7	56.0	2.70x
Average power (W)	22.77	20.40	0.90x

	Llama-3.2-1B	Llama-3.2-3B	3B / 1B
Decode speed (tok/s)	26.2	9.9	0.38x
Working function (harness / re-scored)	5 / 5 of 5	5 / 5 of 5	-
Energy vs. output tokens fit (slope, R ²)	1.057 J/tok, 0.997	2.272 J/tok, 0.716	-

Energy per output token here includes the fixed overhead spread over the output, so it is higher than the decode marginal cost in 1.3. With only 5 runs per model the fit in the last row is indicative only.

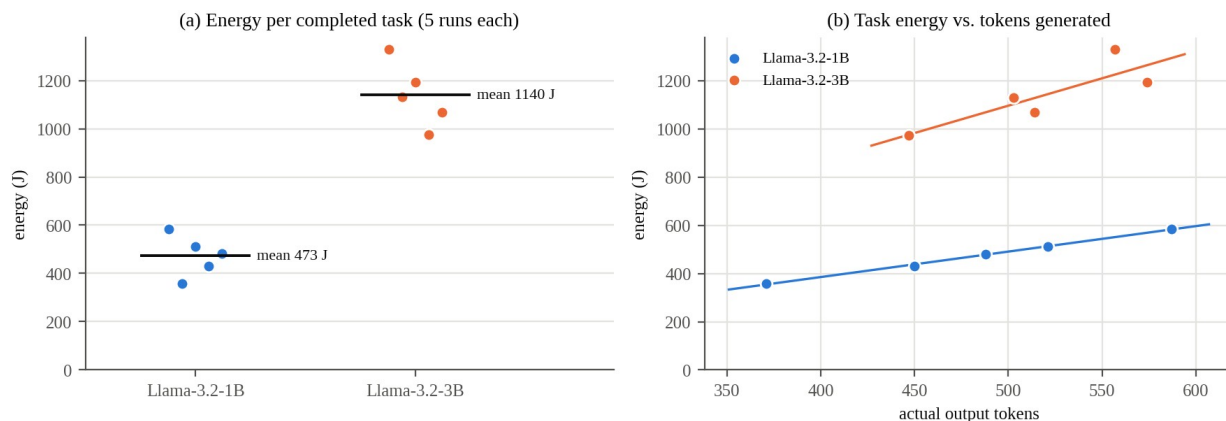


Figure D1. Energy per completed task, individual runs and mean (a), and task energy vs. number of tokens generated with per-model linear fit (b).

4. Cross-Model Summary

metric	Llama-3.2-1B	Llama-3.2-3B	3B / 1B
Parameters (approx.)	1.2 B	3.2 B	2.6x
Decode marginal cost (J/token)	0.875	1.880	2.15x
Prefill marginal cost (J/token)	0.238	0.536	2.26x
Constant overhead per run (J)	131	305	2.32x
Decode speed at 104 / 1693 input tokens (tok/s)	27.0 / 16.3	10.6 / 6.1	0.39x / 0.37x
Average power, Exp 1 output sweep (W)	22.6	18.4	0.81x
Exp 2 mean energy per 256-token run (J)	373.0	825.4	2.21x
Exp 2 working functions, of 20 (harness / re-scored)	8 / 19	8 / 16	-
Exp 3 energy per completed task (J)	473.4	1139.7	2.41x
Exp 3 working functions, of 5 (harness / re-scored)	5 / 5	5 / 5	-

Data notes

- Code checks are shown twice: as recorded by the harness during the runs (scripts from commit 7d8b567, unchanged through eed365e), and re-scored from the saved outputs with the fixed checker (scripts/rescore_code.py -> *_rescored.csv, column passes_tests_v2; originals untouched). The harness failed every run whose output ended mid example-usage line even when the function was complete, e.g. all 5 greedy runs per model. Exp 3 is unaffected (identical in both).
- The llama_cpp_commit field in these runs' meta.txt files is the fyp repo commit at the time (c18adb8 for 1B, 0d96dbc for 3B), not the llama.cpp build: the harness picked up the enclosing git repo. Both models ran the same llama.cpp binary path.
- Week 3 vs. week 5 (Llama-3.2-1B): decode slope 0.76 -> 0.87 J/token and overhead 168 -> 131 J. Week 5 has more points and 5 repeats, and extracted text is now stripped of carriage returns (slightly changes token counts), so treat direct week 3 comparisons with care.